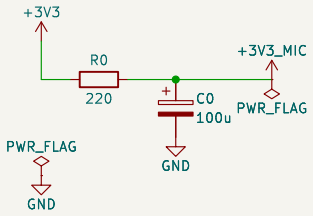
Preamp supply filter



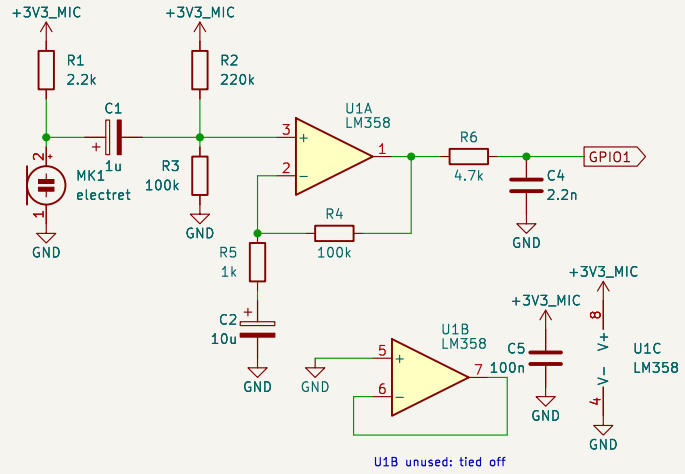
Board 3V3 -> R0/C0 -> +3V3_MIC (preamp only).

ESP32-S3 (U2) pin use

Net	U2 pin	Goes to
+3V3	3V3	R0
+5V	5V (USB)	J1 VCC
GND	GND x2	preamp, J1
GPIO1	GPIO1 ADC1_CH0	preamp out
GPIO11	GPIO11 SPI2 MOSI	J1 DIN
GPIO12	GPIO12 SPI2 SCLK	J1 CLK
GPIO10	GPIO10 CS	J1 CS
GPIO14	GPIO14 pull-up	SW1 MODE
GPIO21	GPIO21 pull-up	SW2 LISTEN

Global labels on this sheet = U2 pins.
Camera ribbon unplugged (Freenove board).

Microphone preamp - LM358 (U1), one stage, gain ~x101

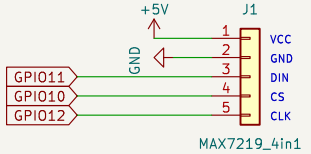


Gain = 1 + R4/R5 = 101 (40 dB): C2 makes the DC gain 1.
Bias = +3V3_MIC x R3/(R2+R3) = 0.95 V. R6 + C4: low-pass ~15 kHz.
On 3.3 V the LM358 output stays below ~1.8 V, so GPIO1 is safe.

Notes

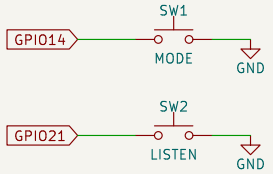
1. All GND symbols are one net. Return the preamp GND and J1 GND to different U2 GND pins.
2. Electrolytics: C0 + to +3V3_MIC, C1 + to MK1, C2 + to R5. MK1 - is the pin joined to its case.
3. DC check before wiring GPIO1: U1 pin 3 ~0.95 V, pin 1 ~0.9-1.1 V (quiet room).
4. Two-stage option (x1100): U1B becomes a x11 stage, see docs/diagrams/preamp-2.png.
5. +3V3_MIC is its own power net (project library spectrum.kicad_sym), not the board 3V3.

LED matrix - 4 x MAX7219, 32 x 8



U3-U6 (MAX7219 + 8x8 LEDs) are on the module.
Known limitation: its ~600 Hz row scan couples into the preamp through GND; run at brightness 0.

Buttons



ESP32-S3 internal pull-ups; pressed = low.

Generated by hardware/kicad/gen_kicad_sch.py (Claude, Anthropic)
One-stage LM358 preamp, ESP32-S3 pins as global labels
Mini-Challenge 2 (CDIO) - Joe Gould

Sheet: /
File: spectrum_visualiser.kicad_sch

Title: Spectrum visualiser - full schematic

Size: A4	Date: 2026-11-06	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1